International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B101

Basic Computer Skills for Digital Forensics

Course Code	CIP-B101
Lab Title	Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis
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1. Introduction

This lab documents a Windows laptop from the perspective of a digital forensic examiner conducting an initial system assessment. The objective was to identify and record key system characteristics, including hardware specifications, operating system details, storage configuration, running processes, and potential locations of digital evidence. Using both native Windows tools and WinHex in read-only mode, the disk's partition structure was examined and partition sizes were independently calculated from sector information to verify storage allocations. The lab also explores the Windows boot process and the file systems present on the machine, providing context for how data is organized and accessed.

2. Tools Used

- Windows 11 Home (Build 26200), HP Laptop 15-da0xxx
- msinfo32, systeminfo, Disk Management, PowerShell (Get-Disk, Get-Partition, Get-Volume, Get-Process), DiskPart
- WinHex, opened on the physical disk in read-only mode
- Kali Linux VM: lsblk, fdisk -l, parted -l, blkid, df -h

PART A

3. System Specification Evidence

This section documents the system's operating system and hardware configuration. The details were collected using the Windows System Information (msinfo32) utility.

Item	Value (from msinfo32 / systeminfo)
OS	Microsoft Windows 11 Home Single Language, 10.0.26200 Build 26200
System type	x64-based PC
Processor	Intel Core i3-8130U @ 2.20 GHz, 2 cores, 4 logical processors
RAM	12.0 GB installed (11.9 GB usable)
BIOS	Insyde F.49, 20/08/2024
BIOS Mode	UEFI
Secure Boot State	On
Boot device	\Device\HarddiskVolume1 (the EFI System Partition)
Storage	WDC WD5000LPCX, 465.76 GB, GPT, Healthy / Online (Get-Disk)

A partition is a logical section of a physical disk, while a volume is a formatted storage area that the operating system can access and use. During the examination, only two partitions with accessible file systems were identified: the C: volume and the Recovery volume. Both were formatted using the NTFS file system.

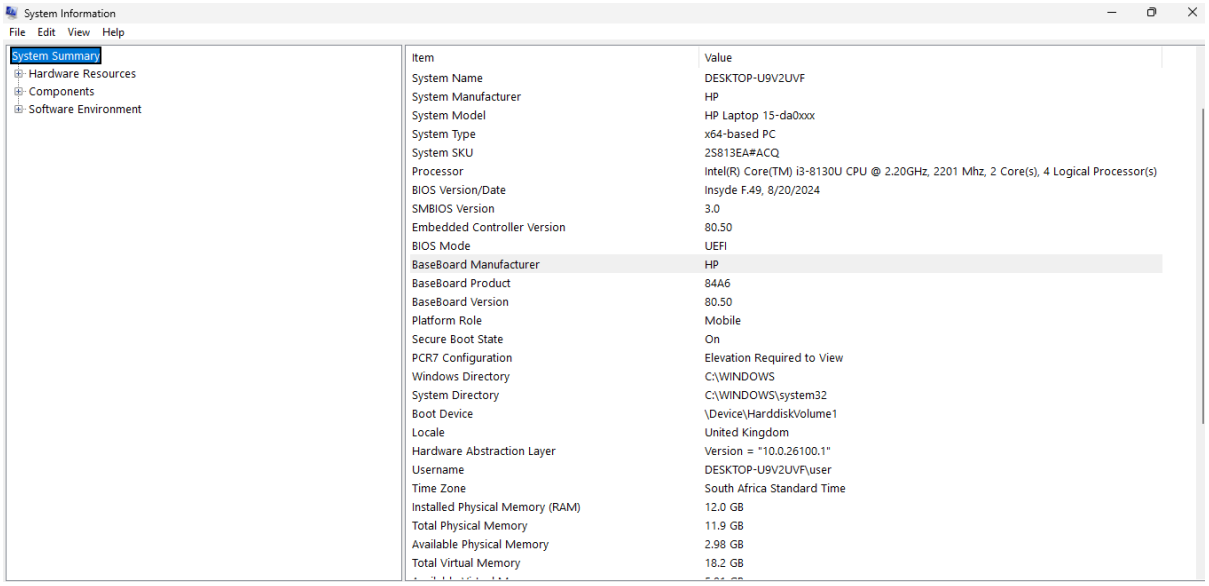


Figure 1: System Information (msinfo32): OS, processor, BIOS Mode = UEFI, Secure Boot State = On, RAM.

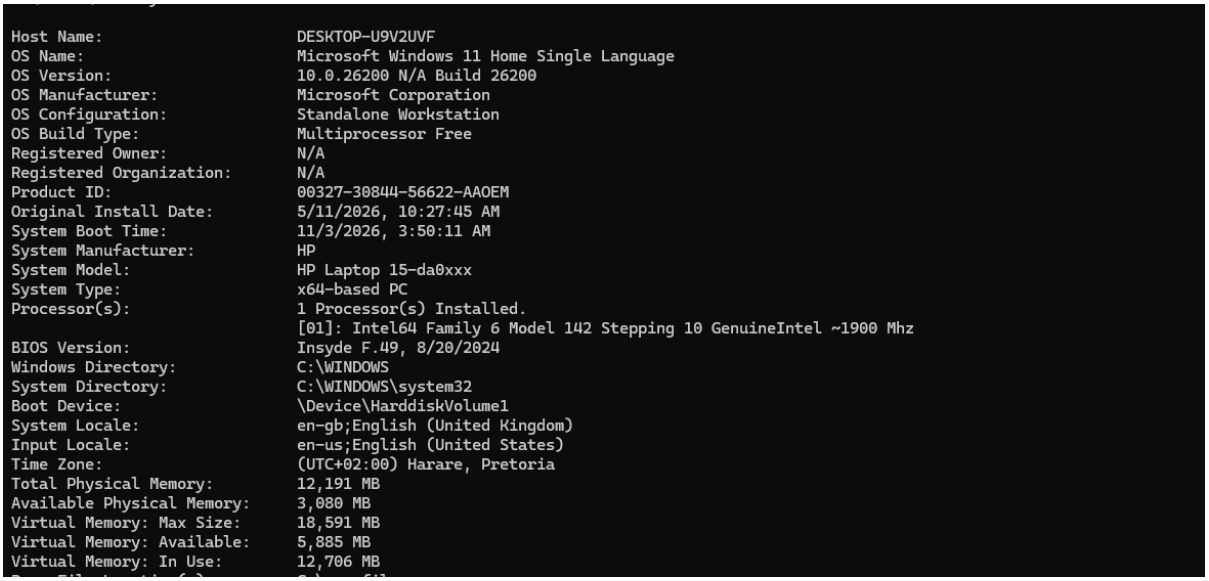


Figure 2: Output of the systeminfo command (first page).

3. Storage information

With Get-Disk, only one Physical disk is identified, disk 0, with 456.76 GB size.

Get-Partition provides more information on the partitions of the disk found, disk 0. The disk has 4 partitions, including partition 3, which is the C drive. The others are the System (contains the boot files), Reserved (for disk management) and Recovery (troubleshooting and repair tools) partitions.

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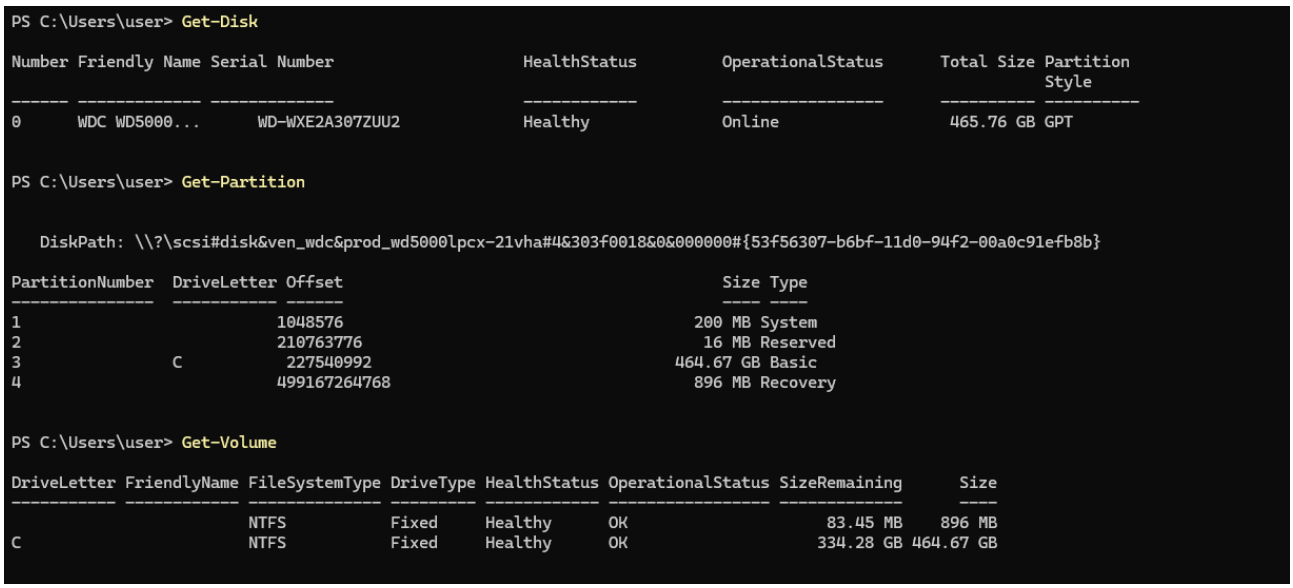


Figure 3: Storage info

4. Evidence Layer Mapping

Layer / source	Example evidence	Where it may be found	Why it matters
User layer	Opened files, search terms, typed usernames	Downloaded files, emails, recent files, browser history	Links a person's actions to system activity
Application layer	Document metadata, browser cache, chat logs	Application folders and databases, application logs	Shows what software created or used the data
OS layer	Event logs, installed programs, user profiles	Windows event logs, registry; Linux /var/log	Records system events including log ins and account activity
Disk layer	Deleted files, fragments, partitions	File system, unallocated space, partition table	Supports recovery and timeline analysis
Memory / process layer	Running processes, network connections, encryption keys, unsaved data, active user sessions	RAM capture, process list	Shows live behaviour that disappears at shutdown
Firmware layer	Boot order, Secure Boot state, boot mode	UEFI settings, EFI System Partition	Explains how the system started and any boot tampering

Why evidence exists in more than one layer

Digital evidence often exists in multiple layers because a single user action can interact with several components of the system at the same time. For example, opening a document may create entries in the application that opened it, generate operating system logs, load data into memory (RAM), and leave traces on the disk. Similarly, visiting a website can leave evidence in the browser history, cache files, DNS records, memory, and network logs. As a result, investigators can often find multiple independent traces of the same activity across different system layers, helping to reconstruct events and verify findings during an investigation. If one trace is deleted, another may survive.

PART B

5. Process Observation

Running processes provide insight into the programs and services that are active on a system at a particular point in time. Processes can reveal user activity, background services, security software, and other applications that may be relevant to an investigation. This section is about programs, processes, and threads by identifying active processes and documenting their purpose and process identifiers (PIDs).

Output of Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet:

```
PS C:\Users\user> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                     Id        CPU WorkingSet
----                     --        --- ----------
agent_ovpnconnect      5612   0.203125   14622720
AggregatorHost         7464              13115392
AppHelperCap           3296              31744000
ApplicationFrameHost  15184   2.109375   46473216
audiodg                9764 198.046875   23498752
chrome                 1452   36.59375   74551296
chrome                 2368   0.890625   52277248
chrome                 3556    1.09375   50012160
chrome                 3596    0.84375   55848960
chrome                 3700    0.71875   48476160
chrome                 3764    0.59375   40693760
chrome                 4012   0.640625   41861120
chrome                 4816   1.328125   57712640
chrome                 5160    2.78125   18972672
chrome                 5860   11.53125   86867968
chrome                 6008    0.78125   40148992
chrome                 6152      0.625   42680320
chrome                 7004        1.5   64782336
chrome                 7076     0.6875   46247936
chrome                 8744   0.671875   45260800
```

Figure 4: Running processes with names, PIDs, CPU time and working set.

Process	PID	Likely purpose
ApplicationFrameHost	15184	Hosts the windows of modern (UWP) apps, ie it interface through which these apps are displayed on the desktop.
audiodg	9764	Windows Audio Device Graph Isolation: processes audio in an isolated process
chrome	1452	One of many Chrome browser processes which could be responsible for a specific tab, rendering or other browser functions.

B2. Program, process and thread

- A **program** is a file stored on disk, a set of instructions that is not running (for example chrome.exe).
- A **process** is a program that has been loaded into memory and is running, with its own process ID, memory space and resources.
- A **thread** is a unit of execution inside a process. A process has one or more threads that share its memory and run in parallel.

Overall, the main difference between them is that a program is the set of instructions on the disk, a process is the running program in memory, and a thread is a task running inside a process process, where one process can have multiple threads and one program can have multiple processes.

Forensic relevance

Process information is valuable because it shows what is actively running on a system at the time of examination. It can help identify suspicious processes, determine which applications were active, and observe how system resources eg CPU are being used.

During malware analysis, process information may reveal malicious programs, unusual parent-child process relationships, unexpected network activity, or processes masquerading as legitimate system files. Malware often runs only in memory and vanishes at shutdown, so recording PIDs, names and parent processes preserves volatile evidence.

During live forensic triage, the process list shows what is running right which helps investigators quickly assess the system's state, identify potential threats, and prioritize areas for further investigation before evidence is lost when the system is powered off.

PART C

6. Disk Sector and Disk Size Calculation

A partition's size can be determined from its sector information. The calculation uses the total number of sectors allocated to the partition and the size of each sector. This provides an independent method of verifying partition sizes and helps to understand how disk space is allocated on the system.

6.1 Sector size and disk capacity

Because sectors are numbered consecutively, the end sector of one partition is equal to the start sector of the next partition minus one.

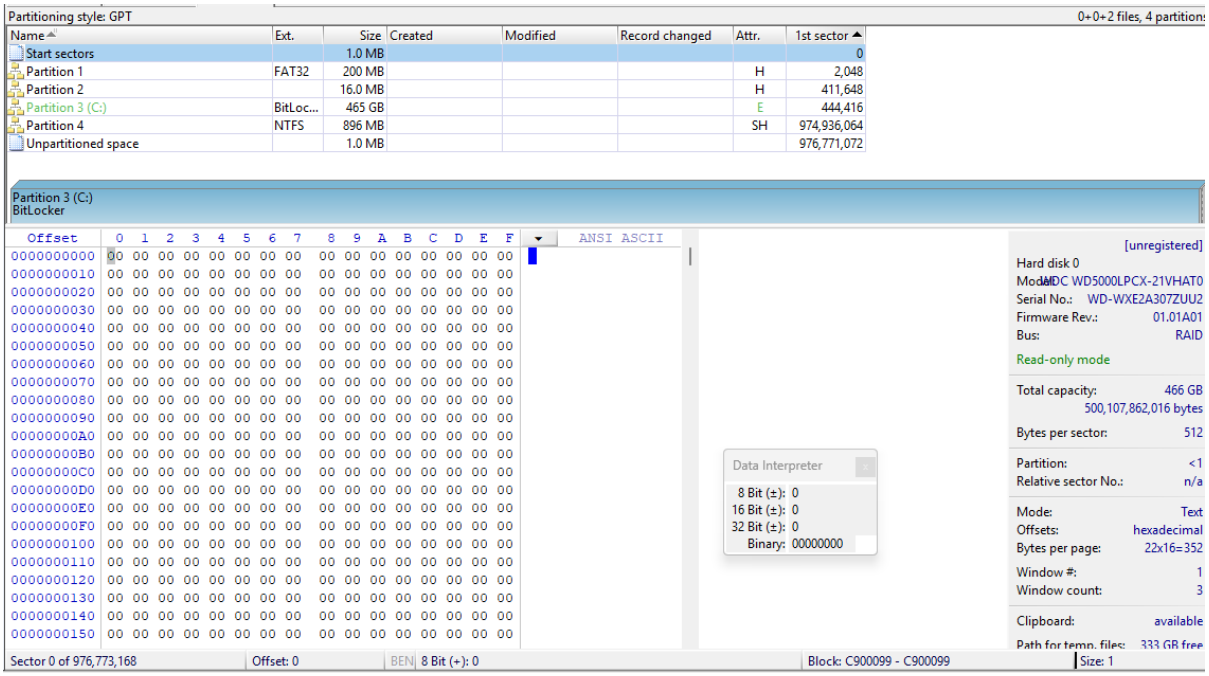


Figure 5: WinHex opened on Hard disk 0 (WDC WD5000LPCX) in read-only mode, showing the partition list and 512 bytes per sector.

```
PS C:\Users\user> Get-Partition -DiskNumber 0 | Format-Table PartitionNumber, Type, Offset, Size

PartitionNumber Type           Offset         Size
--------------- ----           ------         ----
              1 System         1048576    209715200
              2 Reserved     210763776     16777216
              3 Basic        227540992 498939723776
              4 Recovery 499167264768    939524096
```

Figure 6: Get-Partition output with byte offsets and sizes.

Partition	start sector	End sector	Number of sectors (last LBA-First LBA +1)	Sector size	File system	Size (in Gib)	Size (decimal GB)
1	2048	411647	409,600	512 bytes	FAT32	0.195	0.210
2	411648	444415	32,768	512 bytes	none	0.016	0.017
3	444416	974936063	974,491,648	512 bytes	Bitlocker	464.67	498.94

Manual Calculation

1. Partition 1: # of sectors= 409600
409600 *512/ 1024 / 1024 / 1024
=0.195 GiB

in GB (decimal)
409600 *512/ 1000 / 1000 / 1000
=0.210 GB

2. Partition 2: # of sectors= 32768
32768 *512/ 1024 / 1024 / 1024
=0.016GiB

in GB (decimal)
32768*512/ 1000 / 1000 / 1000
=0.017 GB

3. Partition 3 :# of sectors= 974491648
974491648 *512/ 1024 / 1024 / 1024
=464.67GiB

in GB (decimal)
974491648 *512/ 1000 / 1000 / 1000
= 498.94 GB

GB and GiB measure digital storage using different multipliers. A gigabyte (GB) uses powers of 10, where 1 GB equals $1{,}000^3$ bytes, while a gibibyte (GiB) uses powers of 2, where 1 GiB equals $1{,}024^3$ (2^{30}) bytes. Since 1 GiB contains more bytes than 1 GB, storage sizes reported in GiB appear smaller than the equivalent size in GB.

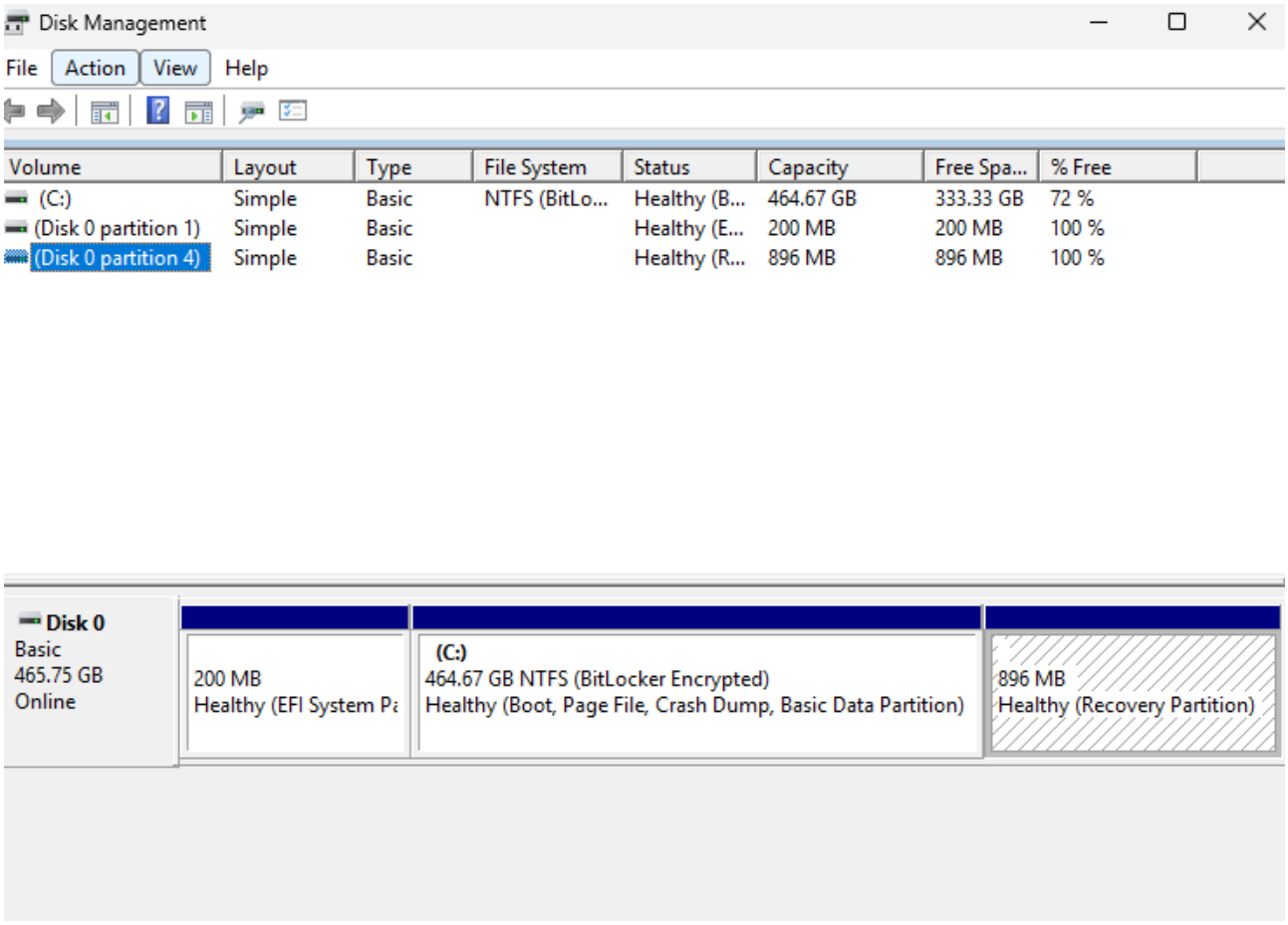
Part D

7. Safe Partition Table Inspection

Disk Management provides a graphical view of the system's storage configuration, including physical disks, partitions, volumes, and unallocated space.

7.1 Using Disk Management

Using diskmgmt.msc, the available volumes and their properties, including capacity and space, were displayed.




```
DISKPART> list disk

  Disk ###  Status         Size     Free     Dyn  Gpt
  --------  -------------  -------  -------  ---  ---
  Disk 0    Online          465 GB  1024 KB        *

DISKPART>
```

DiskPart: showing Disk 0

```
DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    System             200 MB  1024 KB
  Partition 2    Reserved            16 MB   201 MB
  Partition 3    Primary            464 GB   217 MB
  Partition 4    Recovery           896 MB   464 GB
```

diskpart showing the columes for the selected disk 0

Disk part confirms the Disk management information including the available physical disk and the partitions. It also shows the offset, ie the location of the start of the partition.

7.2 Using WinHex : Confirming uses GPT

WinHex was opened in read-only mode to examine the disk structure without modifying any data. Analysis of the partition table showed that the disk uses the GUID Partition Table (GPT) scheme, which is the modern partitioning standard used on UEFI-based systems

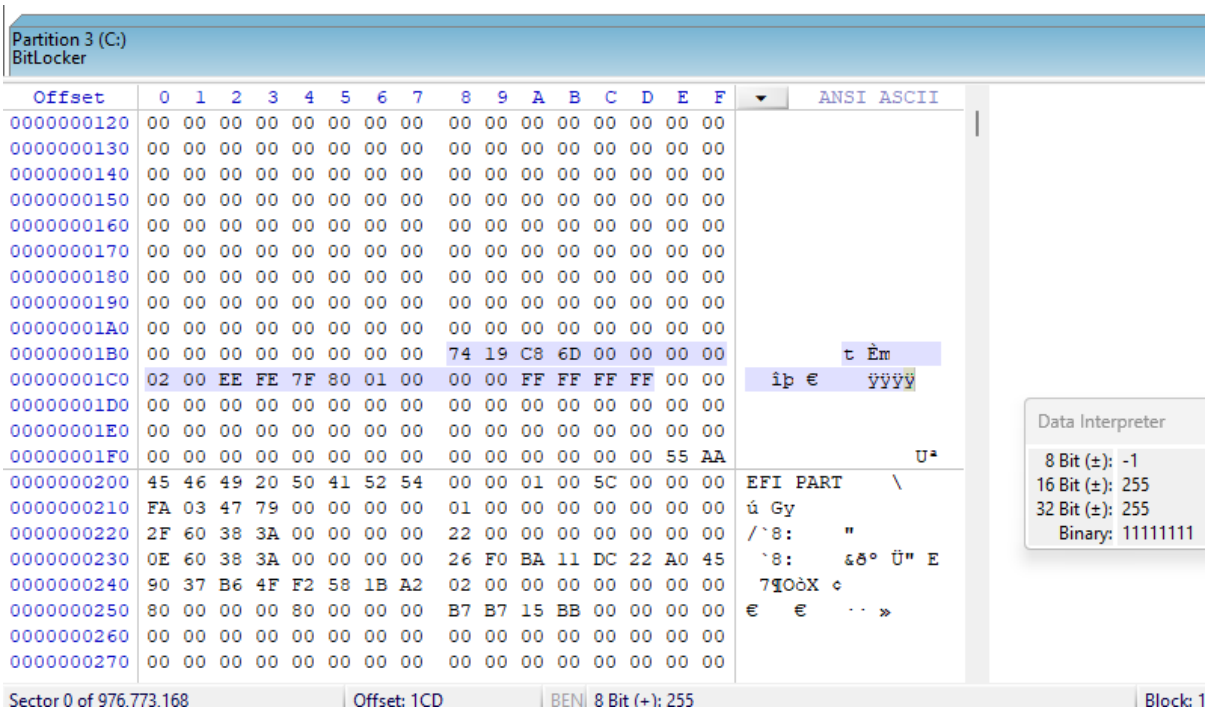


Figure 6: WinHex sector 0 (protective MBR: type EE at 0x1C2, signature 55 AA) followed by sector 1 (GPT header "EFI PART").

Sector 0 (protective MBR):

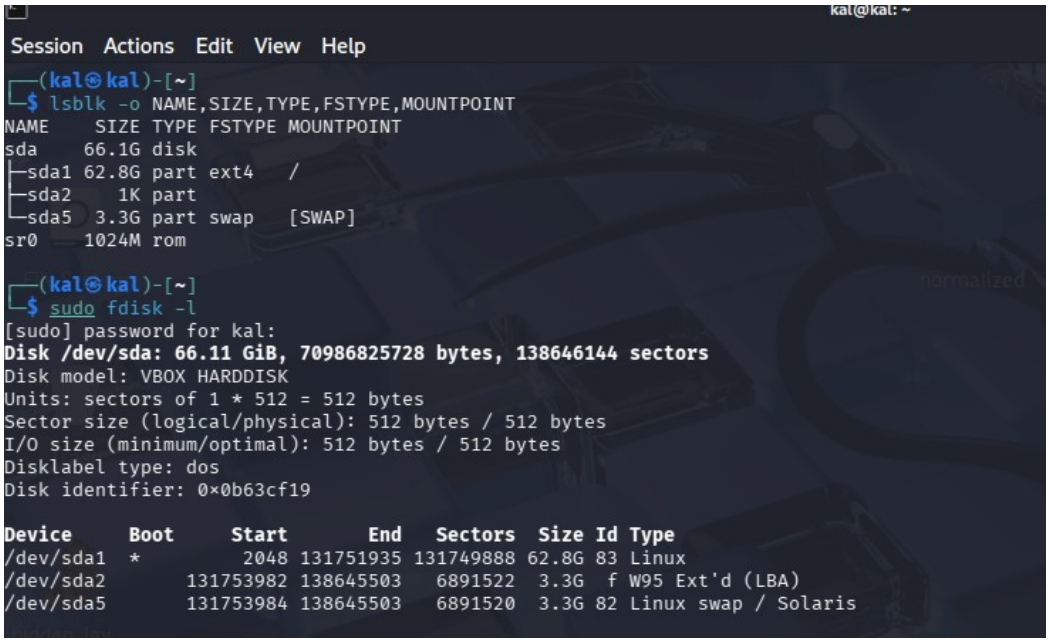
- Row 0x1C0 reads 02 00 EE FE 7F 80 01 00 00 00 FF FF FF FF.
- The EE byte is the GPT protective partition type. The start LBA (0x1C6) is 01 00 00 00 = 1, which points at the GPT header.

Sector 1 (GPT header):

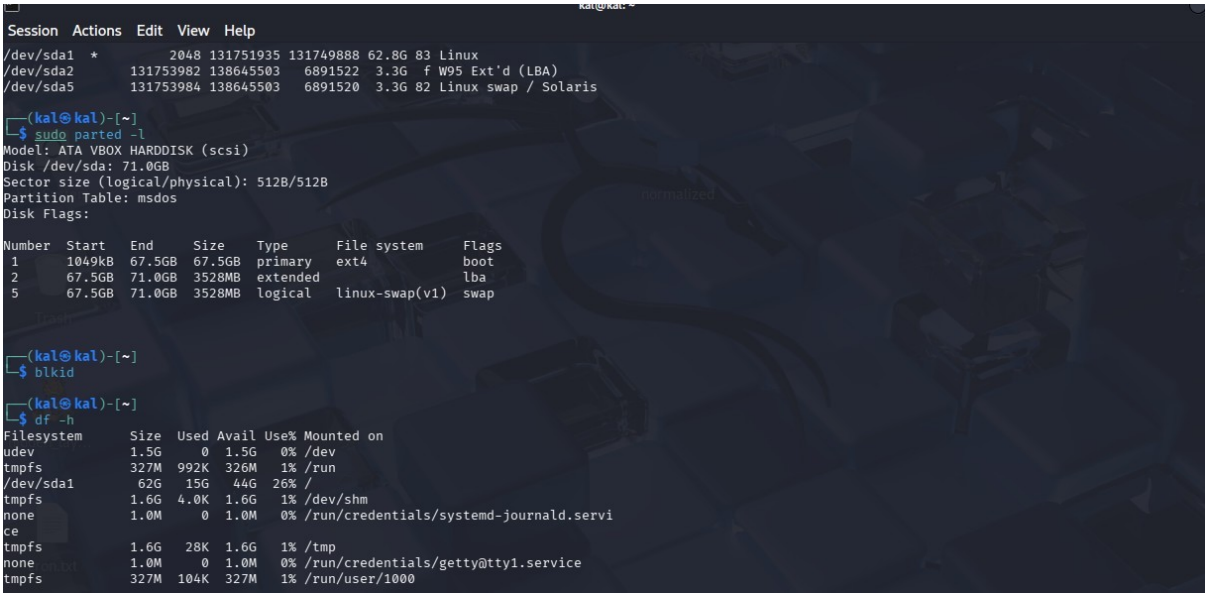
- Starts with 45 46 49 20 50 41 52 54 = "EFI PART" in ASCII.

Therefore, a conclusion was made that he disk uses GPT, with a protective MBR in sector 0.

7.3 Supplementary: Kali Linux VM



Kali: lsblk and fdisk -l output (66.11 GiB virtual disk, DOS label).



Kali: parted -l, blkid and df -h output.

The fdisk -l output showed a 66.11 GiB virtual disk using the DOS/MBR partitioning scheme. The main partition, /dev/sda1, spans from sector 2,048 to 131,751,935, giving 131,749,888 sectors and a calculated size of 62.82 GiB, which matches the 62.8G reported by lsblk. The /dev/sda5 swap partition spans 6,891,520 sectors, resulting in a size of 3.29 GiB. These calculations match the values reported by both fdisk and lsblk, confirming the accuracy of the partition information.

Part D

8. Boot Process Explanation

1. **Power-on:** the power supply stabilises and the CPU starts executing firmware code.
2. **POST (Power-On Self-Test):** the firmware checks the CPU, memory and basic hardware.

3. **UEFI firmware (the successor to BIOS):** firmware settings stored in CMOS are loaded, including boot order and Secure Boot.
4. **GPT / EFI boot record:** UEFI reads the GPT, finds the EFI System Partition (FAT32) and loads the boot entry (bootmgfw.efi on Windows). On a legacy BIOS/MBR machine, the BIOS would read the MBR in sector 0 and run its boot code instead.
5. **Boot loader / manager:** Windows Boot Manager selects the OS and loads the Windows Boot Loader
6. **Kernel loading:** the Windows kernel and drivers are loaded.
7. **OS control:** services and the user session start, and logs and user activity begin to accumulate.

On my system, msinfo32 shows **BIOS Mode = UEFI** and **Secure Boot State = On** (Figure 1). This is supported by partition table evidence: a GPT disk, a protective MBR (EE), an EFI System Partition.

MBR vs GPT

Feature	MBR	GPT
Design	1980s; one table in sector 0 with the boot code	Modern, used with UEFI; header at LBA 1
Size limit	About 2 TB (32-bit LBAs)	Very large (64-bit LBAs; the slide notes about 9.4 ZB)
Partitions	4 primary	Usually 128 entries
Redundancy	None	Primary and backup header and entries
Forensic issue	Damage to sector 0 breaks interpretation	Backup copy at the end of the disk helps recovery and validation

PART F

9. File System Identification Part F

```
PS C:\Users\user> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    83.45 MB    896 MB
C                        NTFS           Fixed     Healthy      OK                   334.28 GB 464.67 GB
```

Get-Volume output: NTFS volumes (896 MB Recovery and 464.67 GB C:).

Partition	File system	Evidence
1 (EFI System)	FAT32	WinHex partition list shows FAT32 (Figure 7)
2 (MSR)	None	Reserved partition, no file system
3 (C:)	NTFS, BitLocker-encrypted	Get-Volume and Disk Management show NTFS (BitLocker Encrypted); WinHex labels it BitLocker
4 (Recovery)	NTFS	Get-Volume, 896 MB; WinHex

Why file systems matter in forensics

File systems are important in digital forensics because they control how data is stored and organized on a disk. By understanding the file system, investigators can locate files, analyze metadata and timestamps, recover

deleted data, and reconstruct user activity. Different file systems store information in different ways, making file system knowledge essential for accurate evidence interpretation. Knowing the file system tells the examiner how to interpret deleted files, unallocated space and timelines.

10. Conclusion

This lab demonstrated how to safely document and profile a computer from a digital forensic perspective without altering any data. The system's hardware and operating system information were identified, running processes were examined, the disk and partition structure were analyzed, partition sizes were verified through sector calculations, and the Windows boot process and file systems were explored. The exercise emphasized the importance of understanding operating systems, storage devices, partitions, and file systems. These concepts are fundamental to digital forensics because they help investigators locate, interpret, and preserve evidence in a reliable and defensible manner. Through this lab, I learned how different components of a computer system contribute to the creation and storage of digital evidence, and why careful analysis are essential during a forensic examination.